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Silicon Photonics Raises New Test Challenges

Semiconductor devices continuously experience advancements leading to technology and innovation leaps, such as we see today for applications in AI high-performance computing for data centers, edge AI devices, electrical vehicles, autonomous driving, mobile phones, and others. Recent technology innovations include Angstrom-scale semiconductor processing nodes, high-bandwidth memory, advanced 2.5D/3D heterogeneous integrated packages, chiplets, and die-to-die-interconnects to name a few. In addition, silicon photonics in a co-packaged optics (CPO) form factor promises to be a key enabling technology in the field of high-speed data communications for high-performance computing applications.

What is CPO?

CPO is a packaging innovation that integrates silicon photonics chips with data center switches or GPU computing devices onto a single substrate (see Figure 1). It addresses the growing demand for interconnects with higher bandwidth and speed, low latency, lower power consumption, and improved efficiency in data transfer for AI data center applications.



Figure 1 Co-packaged Optics (Source: Broadcom)

To understand CPO we need to first understand its constituent technologies. One such critical technology for CPO is silicon photonics. Silicon photonics provides the foundational technology for integrating high-speed optical functions directly into silicon chips. CMOS foundries have developed advanced processes based on silicon semiconductor technology to enable photonic functionality on silicon wafers. CPO uses heterogeneous integrated packaging (HIP) that integrates these silicon photonics chips directly with electronic chips, such as AI accelerator chips or a switch ASIC, on a single substrate or package. Together, silicon photonics and HIP deliver CPO products. Thus, CPO is the convergence of silicon photonics, ASICs and advanced heterogeneous packaging capability supply chains.

As mentioned earlier, CPO brings high-speed, high bandwidth, low latency, low-power photonic interconnects to the computation beachfront. In addition, photonics devices are almost loss-less for large distances enabling one such AI accelerator to share workloads with another AI accelerator hundreds of meters away, while acting as one compute resource. This high-speed and long-distance interconnect CPO fabric promises to re-architect the data center, a key innovation to unlock future AI applications.

Early CPO prototypes are being developed as of 2025 which integrate photonics “engines” with the switch or GPU ASICs on a single substrate, rather than using advanced heterogeneous packages for integration. The optical “engine” in this context refers to the packaging of the silicon photonics chips with other discrete components plus the optical fiber connector; and CPO in this context refers to the assembly of several optical engines with the switch or GPU ASICs on a common substrate.

How to Shorten CPO Time to Market?

The datacom market for CPO presents an opportunity size of about two orders of magnitude higher than what silicon photonics manufacturing supply chains have been historically accustomed to handling, such as the high-mix, low-volume products and applications in telecom and biotech. To successfully achieve CPO at this higher volume, three elements need to advance:

1. The silicon photonics supply chain needs to scale up capacity and achieve high yields at wafer and at the optical engine level.
2. New heterogeneous integrated packaging concepts need to be proven with the OSATs and contract manufacturers for co-packaged optics.
3. New, high-volume test techniques need to be developed and proven, as the current silicon photonics testing processes are highly manual and not scalable for high volume manufacturing.

CPO technology is not mature or at high volume yet, but test equipment providers and device suppliers need to be ready for its arrival as it has a direct impact on automated test equipment test requirements, whether at the wafer, package, or system level. Investments in photonics testing capabilities are critical for developing hybrid testing systems that can keep pace with rapid advancements in photonics, and can handle both electrical and optical signals simultaneously. CPO testing requires active thermal management, high power, large package handling, custom photonic handling and alignment, high-speed digital signaling, wide-band photonic signaling, and high frequency RF signal testing.

Additionally, there are multiple test insertions from wafer to final package test that need to be optimized for test coverage, test time, and cost (see Figure 2). Expertise and experience are required to optimize the test coverage at each insertion to avoid incurring significant product manufacturing cost in both operational expense and capital equipment.



Figure 2: Silicon Photonics Wafer-to-CPO Test Insertions

CPO Test Challenges

Testing CPO devices presents unique challenges due to the diverse processes and materials involved, both electrical and photonics. A unique challenge lies in the inherent complexity of aligning optical components with the precision needed to ensure reliable test results. Optical signals are highly sensitive to minute deviations in alignment, unlike traditional electrical signals, where connection tolerances are more forgiving. The intricacies of CPO, which integrate photonics with high-digital content computing devices, demand precise positioning of lasers, waveguides, and photodetectors. Even the smallest misalignment can result in signal degradation, power loss, or inaccurate measurements, complicating test processes. As this technology advances, automated test equipment needs to evolve to accommodate the precise requirements posed by photonics and optical-electrical integration.

In addition to the precision required, the materials and processes involved in CPO introduce variability. When multiple optical chiplets from different suppliers, each using possibly different materials or designs, are packaged into a single substrate, maintaining alignment across these disparate elements becomes exponentially more difficult. Each optical chiplet may have its own unique optical properties, meaning that test equipment must handle a range of optical alignments without compromising the accuracy of signal transmission or reception. This increases the demand for automated test equipment to adapt and provide consistently reliable measurements across various types of materials and optical designs.

The time-consuming nature of achieving precise alignment also creates a significant bottleneck in high-volume semiconductor testing environments. Aligning optical components, often manually or through semi-automated processes, adds time to test cycles, which can negatively impact throughput and efficiency in production environments. To mitigate these delays, automated test equipment suppliers must invest in advanced photonics testing capabilities, such as hybrid systems that can handle both electrical and optical signals simultaneously and efficiently. These systems must also incorporate faster, more reliable alignment techniques, potentially leveraging AI-driven calibration and adaptive algorithms that can adjust in real-time.

Test meets the high-stakes needs of CPO

With the push for faster data interconnects supporting the latest industry protocols—such as PCIe 5.0/6.0/7.0 and 400G/800G/1.6T b/s Ethernet, and beyond—the stakes are high for data center reliability and performance. Any failure or suboptimal performance in data interconnects can lead to significant downtimes and performance bottlenecks. Consequently, there is a greater emphasis on enhanced test coverage to identify and address potential issues before the components are deployed in data centers. As a result, the semiconductor test industry must provide comprehensive test solutions that cover all aspects of component performance, including signal integrity, thermal behavior, and power consumption under a range of operating conditions.

Ultimately, the industry's shift toward CPO will demand a transformation in test methodologies and equipment, with special emphasis on accurate optical alignment at all test insertions, from wafer to CPO packages. Semiconductor test leaders who invest in advanced photonics testing systems will be better positioned to handle the complexities of this emerging technology, ensuring that they can keep pace with both rapid advancements and growing market demands.

Teradyne is at the forefront of these innovations anticipating new technologies and taking a proactive approach to developing flexible and effective automated test equipment capabilities for the latest advancements in semiconductor packaging and materials.



Dr. George S. Hurtarte is currently Senior Director of SoC Product Strategy in the Semiconductor Test group at Teradyne. George has held various technical, management and executive positions at Teradyne, Lam Research, LitePoint, TranSwitch, and Rockwell Semiconductors. George is in the Advisory Board of SEMI of North America and serves as co-chair of the IEEE Heterogeneous Integration Roadmap (HIR) Test Chapter. George holds a PhD in Electrical Engineering, and three master's degrees (MBA, Computer Science, and Telecommunications). He is also visiting professor at the University of California, Santa Cruz and at the University of Phoenix. He is co-author of the book Understanding Fabless IC Technology.

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